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Gold(I) Catalysis

Room Temperature Allenation of Terminal Alkynes with Aldehydes

Junzhe Xiao, Yifan Cui[+], Can Li[+], Haibo Xu[+], Yizhan Zhai[+], Xue Zhang, and Shengming Ma**

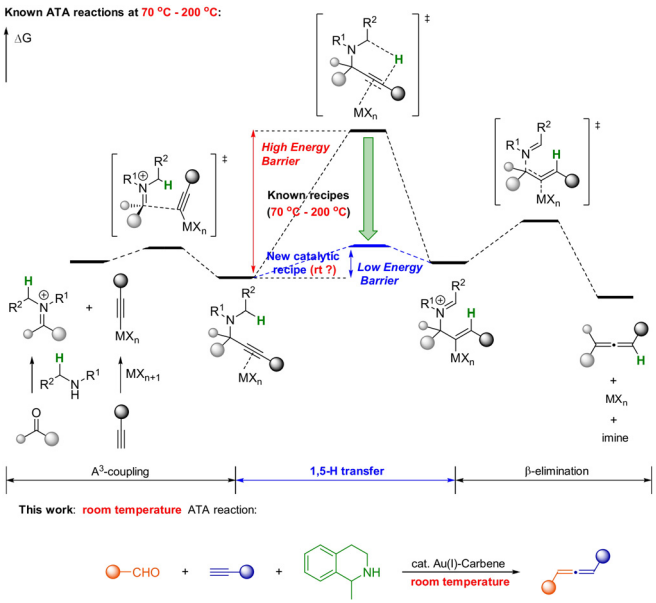
In memory of Dr. Ei-ichi Negishi

Abstract: *A gold-catalyzed room temperature allenation of terminal alkynes (ATA) with aldehydes affording 1,3-disubstituted allenes with diverse functional groups has been developed by identifying a gold(I) catalyst and an amine. The practicality of this reaction has been demonstrated by a ten gram-scale synthesis and the synthetic potentials have been demonstrated via various transformations and formal total synthesis of (−)-centrolobine. Mechanistic studies revealed that the gold catalyst, the aldehyde effect, the fluoroalkyl hydroxyl solvent (TFE or HFIP) and the structure of amine are vital in this room temperature ATA reaction.*

Chemistry of allenes has been intensively investigated and these elegant three-carbon units have been found to be very powerful in organic transformations due to their unique orthogonal cumulative π-systems,[1,2] thus, development of efficient methods for allene syntheses are of high current interest. Pioneered by Crabbé with paraformaldehyde and diisopropylamine,[3a] transition metal-mediated allenation of terminal alkynes (ATA) reaction has been developed for such a purpose with the scope being extended to normal aldehydes and ketones.[3] However, there are two challenges: (1) The reaction consists two steps: A^3-coupling reaction affording propargylic amines and the subsequent 1,5-H transfer-β-elimination. The rate-determining 1,5-H transfer step is of a very high energy barrier,[4] usually requiring reaction temperatures of above 100 °C, which is problematic in many cases for the relatively sensitive allene syntheses (Scheme 1). (2) In most cases the reaction is not catalytic in metal due to the intrinsic nature of in situ formed water and imine, which must have reduced the catalytic performance of the metal species. Inspired by the very limited protocols at 70 °C by Hashmi[3h] and us,[3g] the gold-catalyzed 1,5-H transfer of propargylic amines,[5] and our own understanding of the mechanism for this transformation, herein, we wish to report a room temperature ATA reaction affording allenes in decent yields with a wide scope by identifying a gold(I)-carbene as the catalyst, 2,2,2-trifluoroethanol (TFE) or 1,1,1,3,3,3-hexafluoroisopropanol (HFIP) as the solvent and 1-methyl-1,2,3,4-tetrahydroisoquinoline as the amine with the 1,5-H transfer being mediated by the aldehyde (Scheme 1).

We started our research with benzaldehyde **1a** and 1-octyne **2a** in the presence of morpholine **3a** and Au(IPr)NTf_2 (Table 1). Interestingly, 18 % yield of allene **4aa** was obtained at 25 °C together with 74 % yield of the intermediate product, propargylic amine **5aaa**. Then the effect of *N*-substituents in the *N*-heterocyclic carbenes (NHCs) gold complexes[6] was screened and IMes, ICy, IAd resulted in lower yields. Gratefully, further modification on the backbone of IPr improved the efficiency: 27 % yield of allene **4aa** and 66 % yield of propargylic amine **5aaa** were obtained by using Au(SIPr)NTf_2. The structure of this catalyst was established by X-ray diffraction.

Next, we turned our attention to the effect of amines (Table 2). Upon replacing morpholine **3a** with piperidine **3b**, the yield of **4aa** was improved to 41 %. Pyrrolidine **3c** gave a yield of 52 % for **4aa**. When we tried 1,2,3,4-tetrahydroquinoline **3d**, no product of **4aa** was observed while the reaction with 1,2,3,4-tetrahydroisoquinoline **3e**[7] afforded **4aa** in 19 %

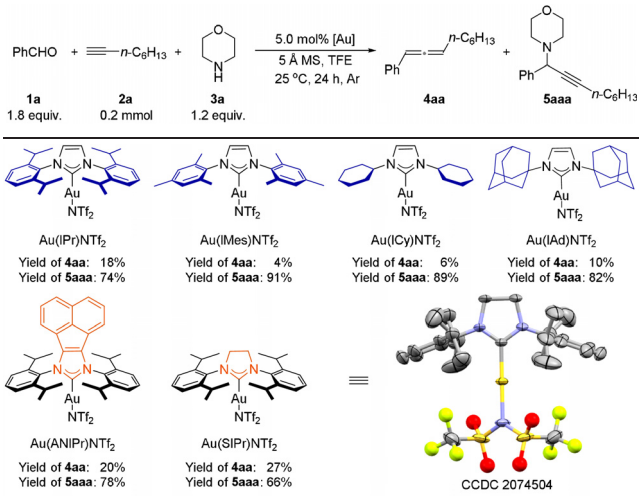


Scheme 1. Au^I-Carbene catalyzed room temperature ATA reaction.

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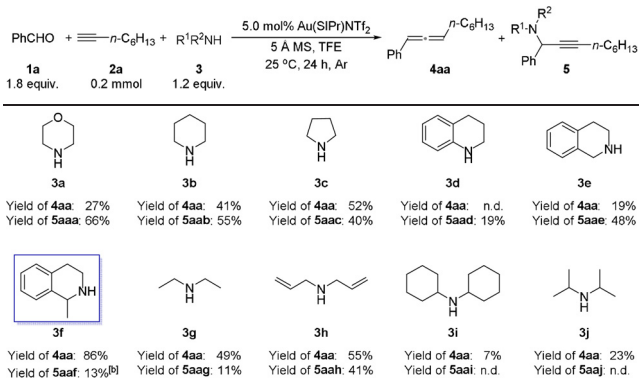
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Table 1: Optimization on Au(NHC)NTf_2[a] and ORTEP representation of Au(SIPr)NTf_2.



[a] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.2 mmol), **3a** (0.24 mmol), [Au] (5.0 mol%), and 5 Å MS in TFE (0.2 mL) at 25 °C for 24 h under Ar atmosphere. Yield of **4aa** and **5aaa** was determined by ^{1}H NMR analysis of the crude product with 2-picoline as the internal standard. TFE = 2,2,2-trifluoroethanol.

Table 2: Screening on R^1R^2NH.[a]

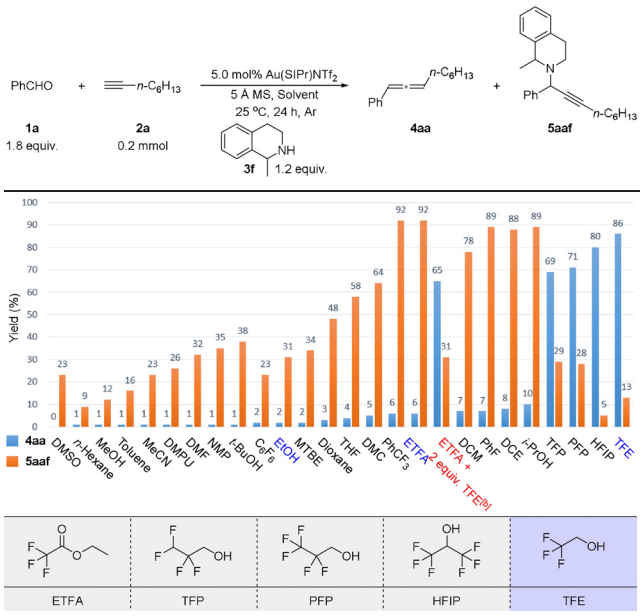


[a] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.2 mmol), **3a** (0.24 mmol), Au(SIPr)NTf_2 (5.0 mol%), and 5 Å MS in TFE (0.2 mL) at 25 °C for 24 h under Ar atmosphere. Yield of **4aa** and **5** was determined by ^{1}H NMR analysis of the crude product with 2-picoline as the internal standard. [b] d.r. = 3.3:1.

yield. Surprisingly, the reaction with 1-methyl-1,2,3,4-tetrahydroisoquinoline **3f** afforded 86% yield of **4aa** with 13% yield of propargylic amine **5aaf**. Further screening on acyclic secondary amines (**3g** to **3j**) did not give rise to better results.

Solvent effects were then carefully studied. The results listed in Table 3 showed that fluoro-substituted alcohols were unique for this room temperature process. Not only did 2,2,2-trifluoroethanol (TFE) give rise to 86% yield of **4aa**, 1,1,1,3,3,3-hexafluoroisopropanol (HFIP) also produced 80% yield of **4aa**. In addition, 2,2,3,3,3-pentafluoropropanol (PFP) and 2,2,3,3-tetrafluoropropanol (TFP) resulted in 71% and 69% yields of **4aa**, respectively. However, other fluoro-substituted solvents without hydroxyl groups, such as C_6F_6,

Table 3: Solvent effect.[a]



[a] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.2 mmol), **3f** (0.24 mmol), Au(SIPr)NTf_2 (5.0 mol%), and 5 Å MS in solvent (0.2 mL) at 25 °C for 24 h under Ar atmosphere. Yields of **4aa** and **5aaf** were determined by ^{1}H NMR analysis of the crude product with 2-picoline as the internal standard. d.r. of **5aaf** range from 1.5:1 to 3.7:1 (see Table S1 in the Supporting Information for details). ETFA = ethyl trifluoroacetate, TFP = 2,2,3,3-tetrafluoropropanol, PFP = 2,2,3,3,3-pentafluoropropanol, HFIP = 1,1,1,3,3,3-hexafluoroisopropanol. [b] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.2 mmol), **3f** (0.24 mmol), Au(SIPr)NTf_2 (5.0 mol%), TFE (0.4 mmol), and 5 Å MS in ETFA (0.2 mL) at 25 °C for 24 h under Ar atmosphere.

$PhCF_3$, ethyl trifluoroacetate (ETFA), PhF, and non-fluoro-substituted alcohols such as MeOH, ^{t}BuOH, EtOH, ^{i}PrOH, afforded **4aa** in less than 10% yields. Other commonly used organic solvents, such as DMSO, *n*-Hexane, Toluene, MeCN, DMPU, DMF, NMP, MTBE, Dioxane, THF, DMC, DCM, DCE, were poor for this reaction. Furthermore, When TFE was compared to ethanol, trifluoromethyl moiety remarkably promoted the A^3-coupling reaction. The ATA reaction with ethyl trifluoroacetate (ETFA) resulted in only 6% yield of allene **4aa** with 92% yield of propargylic amine **5aaf**. However, when 2 equiv of 2,2,2-trifluoroethanol were added, the yield of allene **4aa** was improved into 65%, indicating that the hydroxyl group must also have mediated the rate-determining 1,5-H transfer step.

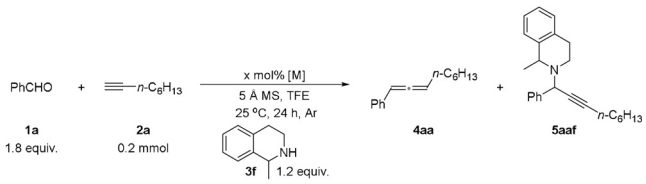
Lastly, Cu, Zn, Cd-based metal salts were examined: the formation of allene **4aa** was essentially not observed with the reaction stopped at the stage of propargylic amine **5aaf**, indicating that gold was irreplaceable in promoting the room temperature 1,5-H transfer (Table 4).

With the optimized condition in hand, we turned our attention to the scope of this room temperature ATA reaction (Table 5). When benzaldehyde **1a** (R^1 = Ph) was used, terminal alkynes with linear alkyl, cyclopropyl, Bn, and Ph substitutions all reacted smoothly to afford allenes **4aa**–**4ae** with good yields. Various functional groups, including halides (**4ba**–**4ea**), methoxyl (**4fa**), cyano (**4ga**), ester (**4ha**), tri-

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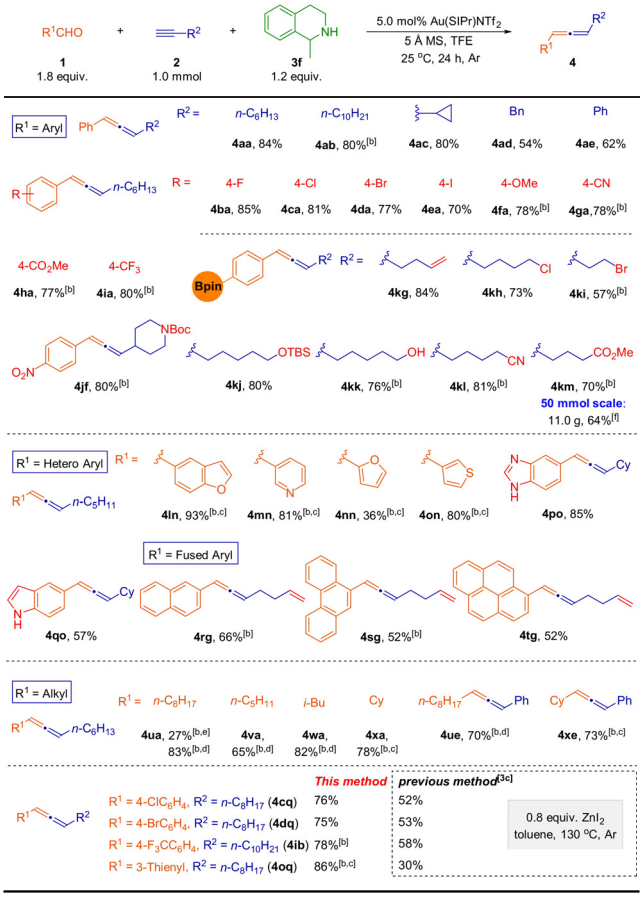
Table 4: Screening on catalyst[a]



Entry	[M] (*x* mol%)	Yield of **4aa** (%)	Yield of **5aaf** (%)
1	Au(SIPr)NTf_2 (5)	86	13 (3.3:1)
2	Cu(SIPr)NTf_2 (5)	n.d.	91 (2.8:1)
3	CuCl (20)	n.d.	62 (5.9:1)
4	CuBr (20)	n.d.	37 (3.6:1)
5	CuI (20)	n.d.	45 (3.5:1)
6	$CuBr_2$ (20)	n.d.	81 (4.4:1)
7	ZnI_2 (80)	n.d.	20 (2.3:1)
8	CdI_2 (80)	1	36 (2.6:1)

[a] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.2 mmol), **3a** (0.24 mmol), [M] (*x* mol%), and 5 Å MS in TFE (0.2 mL) at 25 °C for 24 h under Ar atmosphere. Yield of **4aa** and **5aaf** was determined by ^{1}H NMR analysis of the crude product with 2-picoline as the internal standard. d.r. of **5aaf** was shown in the parentheses.

Table 5: The Scope Studies.[a]



[a] The reaction was conducted with **1** (1.8 mmol), **2** (1.0 mmol), **3f** (1.2 mmol), Au(SIPr)NTf_2 (5.0 mol%), and 5 Å MS in TFE (1.0 mL) at 25 °C for 24 h under Ar atmosphere. Isolated yield. [b] 48 h. [c] 35 °C. [d] 60 °C. [e] Yield was determined by ^{1}H NMR analysis of the crude product with CH_3NO_2 as the internal standard. [f] The reaction was conducted with **1k** (90 mmol), **2m** (50 mmol), **3f** (60 mmol), Au(SIPr)NTf_2 (2.5 mol%), and 5 Å MS in TFE (50 mL) at 25 °C for 48 h under Ar atmosphere to afford 11.0 g of **4km**.

fluoromethyl (**4ia**), and nitro (**4jf**) groups, were well tolerated within aromatic aldehyde **1**. To our delight, aryl aldehyde may also accommodate useful boronate unit (**1k**) and functionalized allenes with vinyl (**4kg**), halides (**4kh**, **4ki**), OTBS (**4kj**), hydroxyl (**4kk**), cyano (**4kl**), and ester (**4km**) groups were afforded in high yields. In addition, hetero-aryl groups (**4ln**–**4qo**) commonly observed in bioactive molecules and fused aryl groups such as naphthyl (**4rg**), phenanthrenyl (**4sg**), and pyrenyl (**4tg**) were also viable. Even the much less reactive aliphatic aldehyde **1u** ($R^1 = n$-C_8H_{17}) produced 27% yield of **4ua** under 25 °C for 48 h. The yield of **4ua** was then improved to 83% by conducting the reaction under 60 °C. A series of aliphatic aldehydes ($R^1 = n$-C_8H_{17}, n-C_5H_{11}, *i*-Bu, Cy) also worked to afford 1,3-dialkyl- and 1-alkyl-3-aryl-disubstituted allenes **4ua**–**4xa** and **4ue**–**4xe**. The practicality was also demonstrated by applying 50 mmol of **2m** ($R^2 = (CH_2)_3CO_2Me$), 90 mmol of **1k** (*R*1 = 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl), 60 mmol of **3f**, and 2.5 mol% Au(SIPr)NTf_2, affording 11.0 g of **4km** in 64% yield under 25 °C for 48 h.

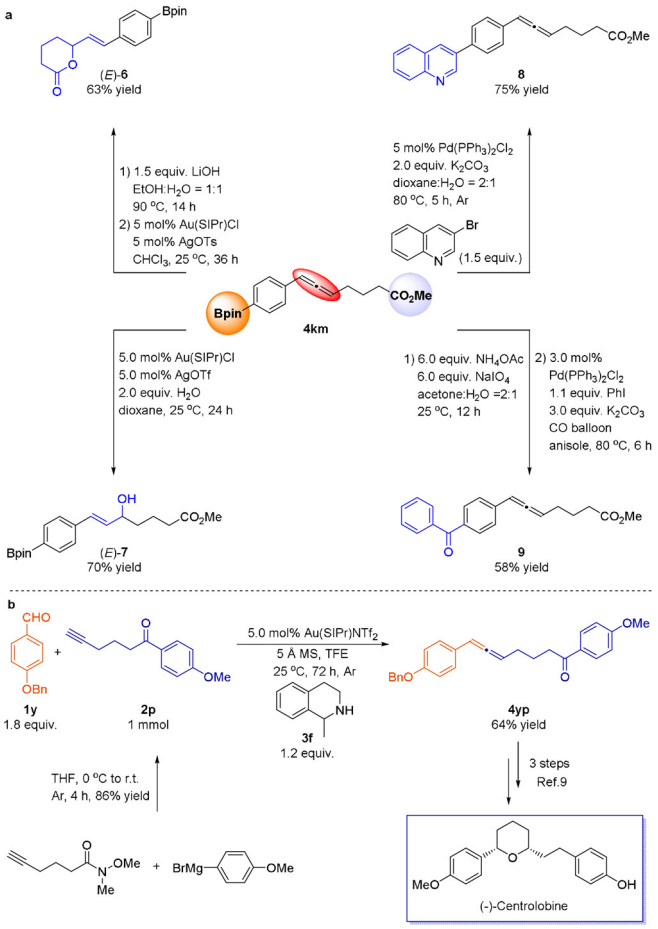
To demonstrate the superiority of this room temperature protocol, we made a comparison between this room temperature ATA reaction and the previous protocol mediated by stoichiometric ZnI_2.[3c] For various useful functional group substituted allenes (**4cq**–**4ib**), this room temperature protocol improved the yields over 20%. Especially, for sensitive heteroaryl aldehyde, the yield of allene **4oq** was improved to 86% from 30% (the last part in Table 5).

Furthermore, synthetic potentials of this reaction have been demonstrated by synthesis of various useful compounds (Scheme 2a). Allene **4km** went through stepwise hydrolysis/gold(I)-catalyzed cycloisomerization[8] to afford δ-valerolactone (*E*)-**6** in 63% overall yield; gold(I)-catalyzed hydration of **4km** selectively formed allylic alcohol (*E*)-**7** bearing useful aryl boronate and ester groups; Suzuki coupling between **4km** and 3-bromoquinoline afforded quinolinyl allene **8** in 75% yield; In the presence of NH_4OAc and $NaIO_4$, **4km** could be transformed to correspond boronic acid, followed by Pd-catalyzed carbonylative coupling with PhI to afford biaryl ketone substituted allene **9** in 58% yield; in addition, the reaction of **1y** and **2p** could produce allene **4yp** in 64% yield, furnishing the formal total synthesis of (−)-centrolobine (Scheme 2b).[9]

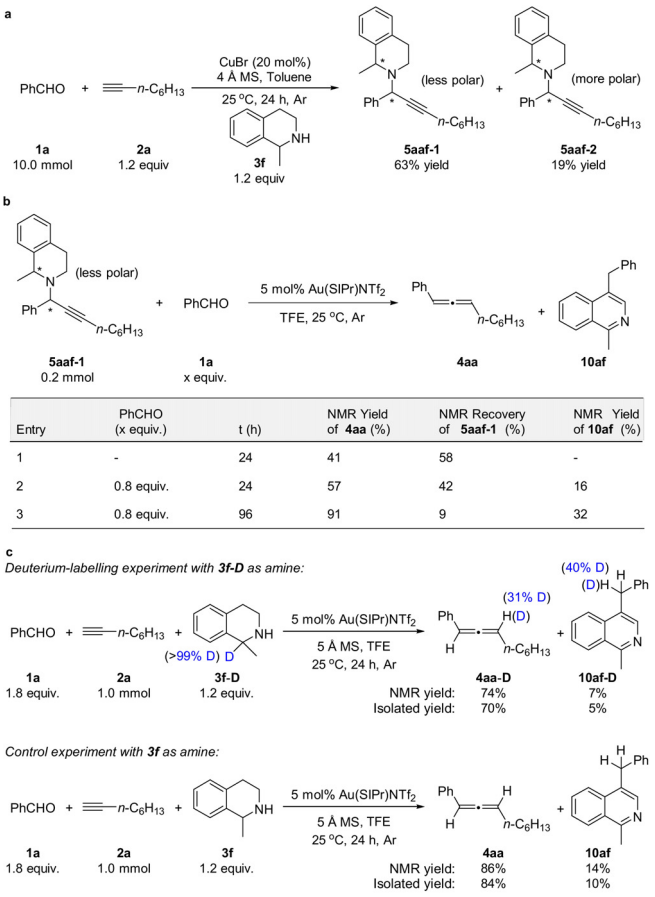
In order to unveil the mechanism, a pair of diastereomers, the less polar propargylic amine **5aaf-1** and the more polar propargylic amine **5aaf-2**, were synthesized via the corresponding A^3-coupling reaction under the catalysis of 20 mol% of CuBr (Scheme 3a).[7] The room temperature 1,5-H transfer of **5aaf-1** was slow with 5 mol% Au(SIPr)NTf_2 (Scheme 3b, Entry 1) and further promoted by excessive 0.8 equiv of benzaldehyde **1a** with the generation of an unexpected byproduct, 4-benzyl-1-methylisoquinoline **10af** (Scheme 3b, Entries 2 and 3), whose structure was identified via X-ray crystallography of **10of**, an analog of **10af** (see the Supporting Information for details). Lastly, a deuterium-labelling experiment applying deuterated amine **3f-D**

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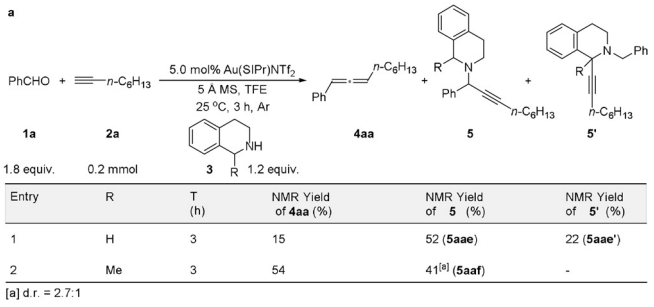
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Scheme 2. Synthetic application. **a.** synthetic utility of bifunctionalized allene **4km. b.** formal synthesis of (−)-Centrolobine.



Scheme 3. Mechanistic studies on 1,5-H transfer.



Scheme 4. Mechanistic studies on amine.

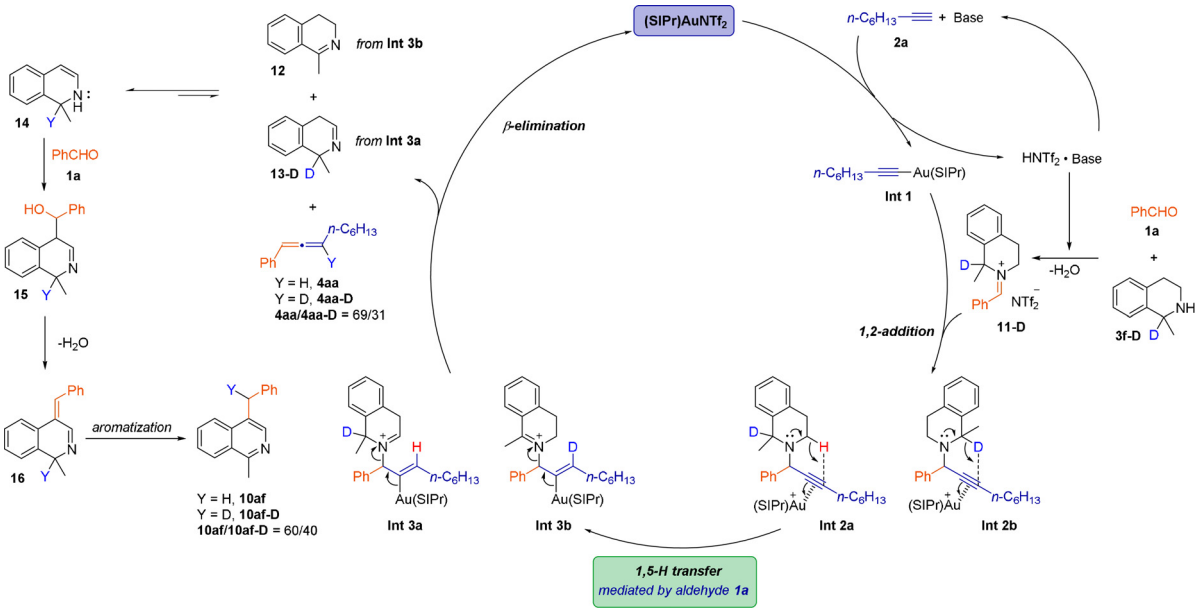
afforded **4aa-D** with 31% of D-incorporation at the 3-position, indicating that in addition to the 1,5-H transfer from 1-position, the same reaction from 3-position of **3f-D** was also viable (Scheme 3c), which also supported by 40% of D-incorporation in the benzylic position of **10af** (**10af-D**).

Control experiment showed that when 1,2,3,4-tetrahydroisoquinoline **3e** was used, the A^3-coupling reaction afforded a regioisomeric mixture of propargylic amines **5aae** and 1-alkynyl 1,2,3,4-tetrahydroisoquinoline derivative **5aae′**[7] in 3 h, while the corresponding **5aaf′** was not generated in the corresponding reaction with 1-methyl-1,2,3,4-tetrahydro-isoquinoline **3f** (Scheme 4a, Entries 1 and 2). Further studies showed that when 1,5-H transfer of **5aae** was conducted in the presence of 0.5 equiv of **5aae′**, the yield of **4aa** dropped to 42% with 54% recovery of **5aae** and 98% recovery of **5aae′**, indicating that the presence of propargylic amine **5aae′** has a negative effect on the 1,5-H transfer of **5aae**. This explains the unique effect of 1-methyl-1,2,3,4-tetrahydroisoquinoline as compared to 1,2,3,4-tetrahydroisoquinoline.

DFT calculations were carried out to show the difference between (SIPr)Au^I and (SIPr)Cu^I on the rate-determining 1,5-H transfer step: For the (SIPr)Au^I-mediated recipe, the transfers of H^a on 3-position and H^b on 1-position takes place with similar activation energies, indicating that both hydrogen atoms may undergo the migration, which agrees well with the deuterium-labelling experiment. Furthermore, (SIPr)Au^I mediated 1,5-H transfer features lower activation energies as compared to the (SIPr)Cu^I-catalyzed system (Figure 1),

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Scheme 5. Proposed catalytic cycle.

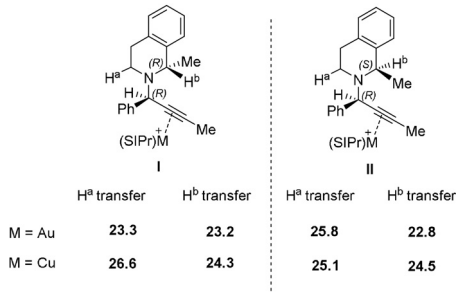


Figure 1. The free energy barriers (ΔG in kcal mol^{-1}) for the 1,5-H transfer of propargylic amine mediated by the (SIPr)Au^{I} or (SIPr)Cu^{I} catalyst.

which is in agreement with observed higher efficiency of (SIPr)Au^{I} over the (SIPr)Cu^{I} catalyst.

Based on these results, we proposed a catalytic cycle[10] as shown in Scheme 5. Alkynyl gold species **Int 1** generated from 1-octyne **2a**, base, and Au(SIPr)NTf$_2$ would undergo 1,2-addition with the in situ generated iminium ion **11-D**. Hydrogen atom on 3-position (**Int 2a**) and deuterium on 1-position (**Int2b**) would transfer to the alkyne moiety to generate alkenyl gold species **Int 3a** and **Int 3b, respectively**, which were somehow mediated by aldehyde **1a** and TFE. β-Elimination of **Int 3a** would afford allene **4aa** and imine **13-D**, while β-elimination of **Int 3b** would afford allene **4aa-D** and imine **12**. Thus, an overall 31 % D-incorporation at 3-position of allene was detected. Enamine **14**, isomerized from imine **12** and **13-D**, would attack the carbonyl group of benzaldehyde **1a** to afford alcohol **15**. Dehydration and aromatization would afford the by-product 1-methyl-4-benzylisoquinoline **10af** with 40 % of D-incorporation at the benzylic position.

In summary, Au(SIPr)NTf$_2$ has been identified for the room temperature gold(I)-catalyzed ATA reaction with a wide scope to synthesize functionalized 1,3-disubstituted allenes with 1-methyl-1,2,3,4-tetrahydroisoquinoline as amine and aldehyde as well as TFE as the mediator. Synthetic potentials have been demonstrated and a formal synthesis of (−)-centrolobine has been realized. Mechanistic study demonstrated that the room temperature 1,5-H transfer must have been promoted by the excess aldehyde and TFE or HFIP. Further studies including the nature of the solvent effect on such ATA reactions are being actively pursued in our laboratory.

Acknowledgements

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Conflict of Interest

We declare no conflict of interest.

Keywords: ATA reaction · gold(I) catalysis · mechanistic study · room temperature

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